

A Comprehensive Survey on Adaptive Continuous User Authentication Using Behavioral Biometrics

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Abstract—

Continuous authentication using behavioral biometrics has emerged as a transformative approach to cybersecurity, enabling identity verification throughout active user sessions rather than only at the point of login. This survey provides a comprehensive analysis of keystroke dynamics, mouse movements, touchscreen gestures, and sensor-based behavioral signals, along with the machine learning and deep learning techniques employed for identity verification. We examine critical challenges including behavioral drift over time, contextual variability across different tasks and environments, device heterogeneity that affects signal consistency, and privacy preservation requirements for real-world deployment. Based on this analysis, we propose a hybrid adaptive authentication architecture that integrates sliding window segmentation, multimodal feature extraction, LightGBM classification, One-Class SVM anomaly detection, and incremental profile updating mechanisms. This architecture is designed to improve long-term usability, robustness against behavioral evolution, and practical deployment capability across diverse application domains including online banking, enterprise security, and mobile platforms.

IndexTerms— Behavioral biometrics, continuous authentication, keystroke dynamics, mouse dynamics, adaptive authentication, deep learning, cybersecurity, multimodal fusion, behavioral drift.

I. INTRODUCTION

Traditional authentication systems, including password-based verification, PIN entry, and even single-factor biometric systems, verify user identity only once at the point of login. While these methods have served as the cornerstone of digital security for decades, they suffer from a fundamental vulnerability: once the initial authentication is complete, the system has no

mechanism to verify that the same legitimate user remains in control of the session. This limitation creates significant security risks including session hijacking attacks, where an attacker takes over an already-authenticated session; insider threats, where a legitimate user's credentials are co-opted by malicious actors; and unauthorized access following device theft after login [7], [13]. The financial impact of these vulnerabilities is substantial, with identity theft losses reaching approximately USD 43 billion annually according to recent industry reports, and phishing attacks increasing by 75% over a two-year period to reach nearly 5 million detected instances.

Continuous authentication systems address this fundamental limitation by monitoring behavioral interaction patterns throughout active sessions rather than relying on a single verification event. These systems collect and analyze how users interact with their devices in real-time, examining characteristics such as typing rhythm (how quickly and with what patterns users press keys), mouse movement trajectories (the paths cursors follow across screens), touchscreen gestures (swipe patterns, tap pressure, and finger contact area), and motion sensor signals (accelerometer and gyroscope readings that capture how users hold and move their devices). The fundamental premise underlying behavioral biometrics is that each individual possesses distinctive, consistent, and difficult-to-replicate patterns of interaction that can serve as a reliable identifier[2], [8]. Unlike passwords that can be stolen or shared, behavioral characteristics are inherently tied to the user and operate transparently without requiring any additional action[11], [13].

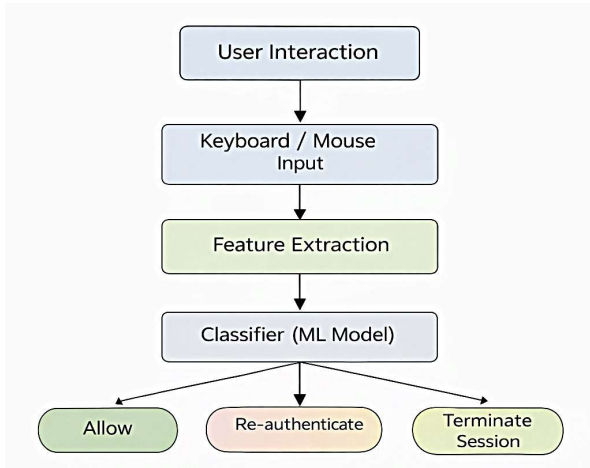


Fig. 1. Continuous behavioral biometric authentication workflow using keyboard and mouse interaction signals.

Behavioral biometrics offers several compelling advantages over traditional authentication methods and even over physiological biometrics such as fingerprints or facial recognition. First, behavioral data can be collected non-intrusively using standard input devices that are already present on computers and mobile devices, eliminating the need for specialized hardware such as fingerprint scanners or iris cameras. Second, authentication occurs transparently in the background without interrupting user workflow, providing security without compromising usability or productivity. Third, behavioral patterns are significantly more difficult to forge or replicate than static credentials, as they involve dynamic characteristics that vary naturally across individuals. Fourth, behavioral biometrics enables continuous monitoring throughout sessions, detecting anomalies even after initial authentication has been performed[5], [8].

Despite these advantages and significant research progress over the past decade, several critical challenges continue to hinder widespread real-world deployment of behavioral biometric authentication systems. Behavioral drift occurs as user interaction patterns naturally evolve over time due to fatigue, skill acquisition through practice, emotional state changes, illness, aging, and adaptation to new devices or software interfaces. A user who types slowly and deliberately when first learning a system may type rapidly and automatically after months of regular use, causing authentication systems trained on early data to incorrectly reject the legitimate user[13], [21]. Contextual variability refers to the observation that the same user exhibits different behavioral patterns when performing different tasks, using different devices, or working in different environments. A user's mouse movements during precise graphic design work differ substantially from their

movements during casual web browsing, and systems that do not account for these contextual differences may generate false rejections[10], [22]. Device heterogeneity means that behavioral signals captured from different keyboards, mice, touchscreens, or smartphones are not directly comparable; typing on a mechanical desktop keyboard produces different timing characteristics than typing on a laptop keyboard or a touchscreen, and authentication systems must handle this variability[8], [21].

Privacy preservation presents another significant challenge, as continuous authentication systems necessarily collect sensitive behavioral data that could potentially reveal personal information about users including their activity patterns, emotional states, health conditions, and even demographic characteristics. Users may be uncomfortable with continuous monitoring, and regulatory frameworks such as GDPR impose strict requirements on the collection, storage, and processing of behavioral data[12]. Finally, the lack of standardized, publicly available long-term datasets spanning months or years makes it difficult to fairly compare different authentication approaches and evaluate their performance under realistic conditions where behavioral drift occurs[7], [13].

This survey provides a comprehensive analysis of the current state of behavioral biometric-based continuous authentication systems. Section II establishes background and fundamental concepts including the distinction between static and continuous authentication, the range of behavioral modalities, and key evaluation metrics. Section III examines keystroke dynamics in depth, covering datasets, feature extraction, and classification approaches from traditional machine learning to deep learning. Section IV explores mouse dynamics and multimodal systems that combine multiple behavioral signals. Section V addresses mobile sensor-based authentication using accelerometers, gyroscopes, and touchscreen data. Section VI examines deep learning approaches including CNN, LSTM, and transformer architectures. Section VII covers privacy-preserving authentication techniques including homomorphic encryption and secure template storage. Section VIII addresses the critical challenge of behavioral drift and adaptive authentication mechanisms. Section IX presents our proposed hybrid adaptive authentication architecture designed to address identified gaps. Section X concludes the survey with recommendations for future research.

II. BACKGROUND AND FUNDAMENTALS OF CONTINUOUS AUTHENTICATION

Continuous authentication represents a paradigm shift from traditional static authentication approaches by verifying user identity repeatedly and transparently throughout an active session rather than only at login time. Unlike conventional authentication mechanisms that rely on passwords, tokens, or one-time biometric verification, continuous authentication systems analyze behavioral interaction patterns dynamically during system usage to ensure that the legitimate user remains in control of the session [8], [13].

Traditional static authentication assumes that the entity authenticated at login remains unchanged throughout the session duration. However, this assumption often fails in real-world environments due to session hijacking attacks, insider threats, and unauthorized access following credential compromise. Continuous authentication addresses these vulnerabilities by monitoring user interaction characteristics such as typing rhythm, mouse movement trajectories, touchscreen gestures, and sensor-based motion signals in real time [7], [22].

Behavioral biometric authentication systems operate transparently in the background without interrupting user workflows, providing a balance between security and usability. Since behavioral interaction patterns are difficult to imitate or reproduce accurately, they offer a reliable mechanism for detecting unauthorized users even after successful login authentication [2], [11].

Despite their advantages, continuous authentication systems must address several challenges including behavioral drift, contextual variability, device heterogeneity, and privacy concerns. These challenges significantly influence the long-term reliability and deployment feasibility of behavioral biometric authentication systems in practical environments [21], [13].

A. Static Versus Continuous Authentication

Static authentication, which encompasses the vast majority of deployed authentication systems today, operates on a simple model: the user provides credentials at the beginning of a session, the system verifies these credentials against stored references, and if verification succeeds, the user is granted access for the duration of the session. This model implicitly assumes that the individual who successfully authenticated remains in control of the session throughout its entire duration. In practice, this assumption is frequently violated. Session hijacking attacks allow attackers to take over authenticated sessions by stealing session cookies or tokens. Insider threats involve legitimate users whose credentials are co-opted by

malicious actors. Device theft after authentication gives attackers direct access to authenticated systems. Credential theft through phishing or keylogging allows attackers to authenticate as legitimate users from different locations. In all these scenarios, static authentication provides no protection because the system has no mechanism to detect that the controlling entity has changed [8], [13].

Continuous authentication addresses this fundamental weakness by performing identity verification repeatedly throughout the session. Rather than relying on a single verification event, continuous authentication systems collect and analyze behavioral data as users naturally interact with their devices [7], [8]. The authentication process operates in the background without requiring explicit user action, extracting features that capture distinctive behavioral patterns and comparing these patterns against stored templates representing the legitimate user's expected behavior [2], [11]. When the observed behavior deviates sufficiently from the expected pattern, the system can take appropriate action ranging from subtle re-authentication challenges (such as requesting the user to re-enter a password or provide a fingerprint) to complete session termination.

The frequency of verification in continuous authentication systems can be adjusted based on risk level [8]. During high-risk activities such as financial transactions or access to sensitive data, more frequent verification can be performed. During low-risk activities such as browsing public information, less frequent verification may suffice. The authentication decision can also incorporate multiple factors: a single anomalous action may not trigger a response, but a sustained pattern of anomalous behavior will.

Several important advantages distinguish continuous authentication from static approaches. First, continuous authentication significantly reduces the attack window available to adversaries who gain access after initial authentication. Research has demonstrated that while an intruder can perform more than 1,000 malicious actions after gaining access to a mobile device using only knowledge-based authentication, multimodal behavioral biometrics reduces this to nearly zero successful actions [5], [8]. Second, continuous authentication provides protection against session hijacking and insider threats that static authentication cannot address. Third, because behavioral biometrics operate transparently, continuous authentication can be implemented without degrading user experience or requiring users to remember additional credentials. Fourth, behavioral characteristics are inherently more difficult to steal or forge than

passwords or physical tokens[13], as they involve dynamic patterns unique to each individual.

However, continuous authentication also presents unique challenges. The requirement for continuous data collection and processing introduces computational overhead and potential battery consumption concerns on mobile devices[7]. The need to make rapid authentication decisions with limited data requires efficient algorithms that can operate reliably with small sample sizes. The inherent variability of human behavior means that authentication decisions must be probabilistic rather than deterministic[[11], requiring careful threshold selection to balance security and usability. Finally, the continuous collection of behavioral data raises privacy concerns that must be addressed through appropriate technical and policy measures[12].

B. Behavioral Biometric Modalities

Behavioral biometrics encompasses a diverse set of modalities, each capturing different aspects of human-computer interaction. The most extensively studied modalities in the context of continuous authentication include keystroke dynamics, mouse dynamics, touchscreen gestures, gait analysis, and various sensor-based signals[2], [8].

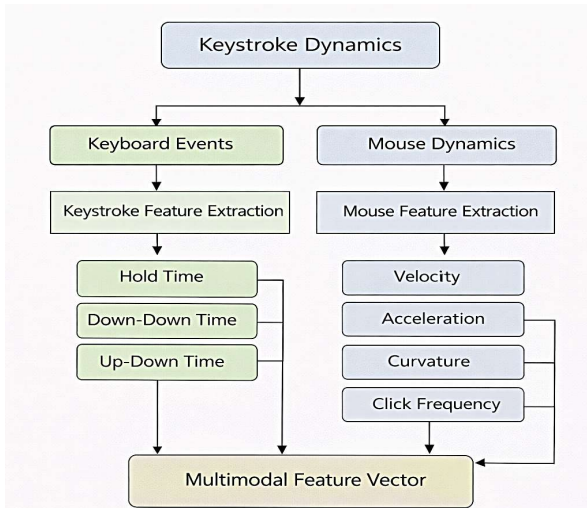


Fig. 2. Multimodal behavioral feature extraction using keystroke timing and mouse trajectory characteristics.

Keystroke dynamics analyzes the timing patterns associated with keyboard typing. When a user types on any keyboard, each key press and release generates precise timing events that can be captured by the operating system or by dedicated logging software. From these raw events, researchers extract various timing features including hold time (the duration between pressing and releasing a single key), flight time (the

interval between releasing one key and pressing the next), down-down time (the interval between pressing consecutive keys), and up-up time (the interval between releasing consecutive keys)[9], [13]. Additional features include typing speed measured in words per minute[6], error rates indicated by backspace and delete key usage, and patterns of special key usage such as Shift, Control, and Alt keys. Keystroke dynamics offers the advantage of being universally applicable across any device with a keyboard, from desktop computers to smartphones with virtual keyboards, and can be collected without specialized hardware. The rich temporal information captured by keystroke timing has been shown to provide strong discriminative power for user identification[11].

Mouse dynamics analyzes the patterns of mouse movement and clicking behavior[1],[10]. When a user operates a mouse, touchpad, or other pointing device, the system captures a continuous stream of events including movement coordinates, button down and up events, and scroll actions. From this raw data, researchers segment the interaction into meaningful units such as point-and-click actions (sequences of movement ending in a click), drag-and-drop actions, and pure movement sequences. Features extracted from these segments include movement speed, acceleration, jerk (rate of change of acceleration)[10], curvature, straightness (ratio of direct distance to actual path length), angle of movement, clicking speed, and clicking strength where measurable. Mouse dynamics is particularly valuable in contexts where typing is infrequent, such as during web browsing, gaming, or interaction with graphical applications, and can serve as a complementary modality to keystroke dynamics in multimodal systems[11].

Touchscreen gestures represent the mobile and tablet analog of mouse dynamics, analyzing the patterns of finger interaction with capacitive touchscreens[7], [8]. Touchscreen data provides richer information than mouse data in several respects, as modern touchscreens can capture not only position and timing but also pressure (the force applied to the screen), area of touch (the size of the finger contact region, which varies with finger orientation and applied force), and multi-touch gestures involving multiple simultaneous finger contacts. Common gestures analyzed for authentication include swipes in various directions, taps, flicks, pinches, spreads, and various custom gestures defined for specific applications[5]. Features extracted from these gestures include velocity, acceleration, path deviation, pressure variation, touch area variation, and gesture duration.

Gait analysis identifies individuals based on their walking patterns using motion sensors embedded in smartphones

and wearables[7], [8]. When a user walks while carrying a mobile device, the device's accelerometer and gyroscope capture characteristic patterns corresponding to the user's stride length, cadence, foot placement, and overall movement dynamics. Gait-based authentication has the unique advantage of operating when the user is not actively interacting with the device, enabling authentication even when the device is in a pocket or bag. This makes gait particularly valuable for continuous authentication in mobile scenarios[5].

Sensor-based signals encompass a wide range of additional modalities enabled by the diverse sensors found on modern mobile devices[7]. These include accelerometer signals capturing device movement and orientation, gyroscope signals capturing angular rotation, magnetometer signals capturing heading and orientation relative to Earth's magnetic field, pressure sensors, proximity sensors, ambient light sensors, and GPS location data. Researchers have explored various combinations of these sensors for authentication, often in conjunction with other modalities[8], [21].

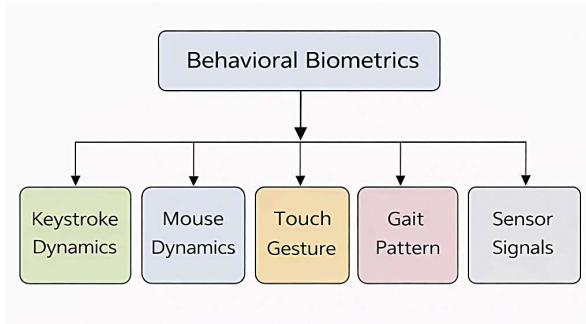


Fig. 3. Classification of behavioral biometric modalities used in continuous authentication systems.

C. Evaluation Metrics for Authentication Systems

The performance of biometric authentication systems is evaluated using several standard metrics that capture different aspects of system behavior[8],[13]. The False Acceptance Rate (FAR) measures the proportion of authentication attempts by impostors that are incorrectly accepted as genuine, calculated as $FP/(FP+TN)$ where FP represents false positives and TN represents true negatives. A low FAR indicates that the system is effective at rejecting impostors, which is critical for security[13]. The False Rejection Rate (FRR) measures the proportion of authentication attempts by genuine users that are incorrectly rejected, calculated as $FN/(FN+TP)$ where FN represents false negatives and TP represents true positives. A low FRR indicates that the system is effective at accepting legitimate users, which is critical for usability. FAR and FRR represent a fundamental trade-off:

tightening the decision threshold to reduce FAR tends to increase FRR, while loosening the threshold to reduce FRR tends to increase FAR.

The Equal Error Rate (EER) is a single-value metric representing the point at which FAR and FRR are equal. The EER is widely used because it provides a threshold-independent summary of system performance, with lower EER values indicating better overall performance[9]. Accuracy measures the overall proportion of correct classifications, calculated as $(TP+TN)/(TP+TN+FP+FN)$. While intuitive, accuracy can be misleading when class distributions are imbalanced. The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings, providing a comprehensive view of classifier performance across the entire operating range[11]. The Area Under the ROC Curve (AUC) provides a single-value summary of ROC performance, with values closer to 1 indicating better discriminative ability[11].

III. KEYSTROKE DYNAMICS

Keystroke dynamics represents the most extensively studied behavioral biometric modality for continuous authentication[2], [13], with research dating back to the pioneering work of Gaines and colleagues in 1980[13]. The fundamental premise is that each individual possesses a distinctive and consistent typing rhythm that can serve as a reliable identifier[6], [9]. This section provides a comprehensive examination of keystroke dynamics-based authentication systems.

A. Types of Keystroke Dynamics Datasets

Keystroke dynamics datasets are generally categorized into static (fixed-text) datasets and dynamic (free-text) datasets[13]. Static datasets collect keystroke data as users type a predetermined text of fixed length and content, typically the same password or phrase across all users and sessions. These datasets are primarily used to evaluate authentication scenarios where the system has advance knowledge of the text being typed[2], such as password-based authentication during login. The CMU Benchmark Dataset[20], collected by Killourhy and Maxion in 2009, is the most widely used static dataset, containing keystroke data from 51 participants who typed the password "tie5Roanl" 50 times per session across 8 sessions, yielding 20,400 samples total[20]. The participant population consisted of 30 males and 21 females with ages ranging from 18-20 to 61-70.

Dynamic datasets collect keystroke data as users engage in unrestricted typing activities that reflect real-world

scenarios[13]. These datasets enable the evaluation of continuous authentication systems that must operate without knowledge of the text being typed. The Buffalo Dataset[9], [15], collected by Sun, Ceker, and Upadhyaya in 2016, contains keystroke data from 148 participants collected over three sessions spanning 28 days, including both transcribed text and free-text responses. The Clarkson II Dataset, collected by Murphy and colleagues in 2017, contains keystroke data from 103 participants typing on desktop keyboards over an extended period of 2.5 years in completely uncontrolled, naturalistic scenarios, making it uniquely valuable for studying long-term behavioral drift[13], [21].

B. Feature Extraction for Keystroke Dynamics

Feature extraction is a critical step in keystroke dynamics-based authentication systems. From raw keystroke events, researchers extract various timing features[6], [13]. Hold time (HT) for a single key is calculated as the difference between release time and press time, capturing how long a user holds down each key[6]. Down-down time (DD) between consecutive keys is calculated as the difference between the press time of the subsequent key and the press time of the current key[6], capturing the overall rhythm of typing. Up-down time (UD), also known as flight time or latency, is calculated as the difference between the press time of the subsequent key and the release time of the current key[6], specifically capturing the gap between keys when no key is pressed. Additional features include typing speed measured in words per minute[9], error rate measured by backspace and delete key frequency, and aggregate statistics including mean, median, standard deviation[13], minimum, and maximum of basic timing features across a typing session

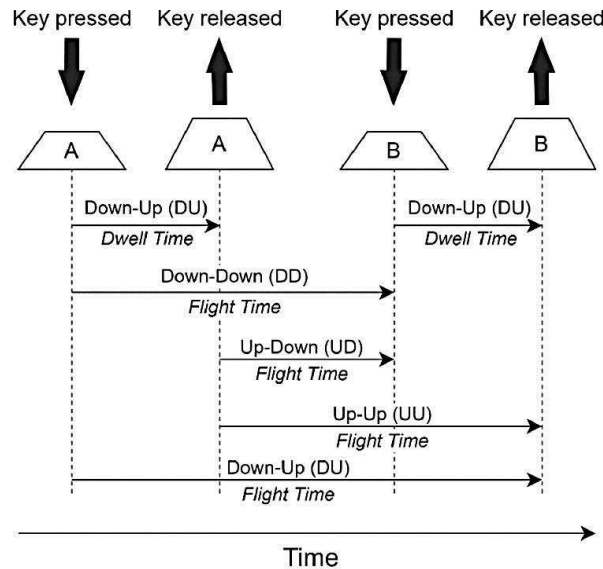


Fig. 4. Keystroke timing feature extraction process showing hold time and inter-key latency measurements..

In addition to feature extraction, feature selection plays a crucial role in reducing dimensionality and improving classifier performance. The Minimum Redundancy Maximum Relevance (mRMR) algorithm is particularly effective for behavioral biometrics, as it selects features that have high correlation with the user identity (maximum relevance) while maintaining low correlation among selected features (minimum redundancy). For a typical keystroke and mouse dynamics system, mRMR reduces the feature space from approximately 100-150 raw features to 30-40 optimized features[6], removing redundant timing measurements and movement characteristics that do not contribute to user discrimination.

C. Classification Approaches for Keystroke Dynamics

Traditional machine learning approaches have been widely applied to keystroke dynamics[11]. Support Vector Machines (SVMs) are effective in high-dimensional spaces and can model complex decision boundaries through kernel functions[6]. Random Forest classifiers provide robustness to noise and overfitting through ensemble averaging[14]. K-Nearest Neighbors (KNN) is simple and requires no training phase but can be computationally expensive at test time[20]. Deep learning approaches have gained substantial popularity in recent years[15]. Multilayer Perceptrons (MLPs) can model complex non-linear relationships but do not explicitly capture sequential structure[11]. Long Short-Term Memory (LSTM) networks are specifically designed to handle sequential data and can capture long-range dependencies in keystroke sequences[9]. Hybrid CNN+RNN architectures use convolutional layers to extract local features followed by recurrent layers to capture longer-range temporal dependencies[15].

Recent research has demonstrated that gradient boosting methods achieve state-of-the-art performance on keystroke authentication tasks[20]. A comparative study by Muralidharan and colleagues found that LightGBM achieved 94.68% accuracy on the CMU dataset[20], significantly outperforming prior hybrid approaches including the POHMM/SVM model which achieved 86.8% accuracy[20].

D. Dataset Collection for Behavioral Authentication Research

In addition to public benchmark datasets, controlled data collection is essential for evaluating authentication systems under specific application scenarios[7]. For

web-based authentication research, data collection is typically performed using a custom web application interface that simulates a banking or enterprise environment[18]. The data collection pipeline follows these steps:

- **User Interaction Recording:** Participants perform structured tasks (e.g., form filling, menu navigation, data entry) while keyboard and mouse events are captured using JavaScript event listeners[10]. Each event records timestamp, key code or mouse coordinates, and action type.
- **Sliding Window Segmentation:** The continuous event stream is divided into overlapping windows of fixed duration (e.g., 60 seconds) with 50% overlap between consecutive windows[14]. Each window generates a feature vector containing statistical summaries of keystroke timing (mean hold time, flight time variance) and mouse movement (average velocity, acceleration, click frequency)[11].
- **Multiple Sessions per User:** To capture natural behavioral variability, data is collected across multiple sessions on different days[13]. Each user contributes 3-5 sessions, with each session lasting 10-15 minutes of active interaction.
- **Impostor Data Collection:** To evaluate authentication performance, cross-user validation is performed where each user acts as an impostor for other users[10]. The dataset is balanced using SMOTE (Synthetic Minority Over-sampling Technique) to address the inherent class imbalance where genuine samples outnumber impostor samples[14].
- **Feature Normalization:** Per-user z-score normalization is applied to account for baseline differences in typing speed and mouse movement characteristics across users[11].

IV. MOUSE DYNAMICS AND MULTIMODAL AUTHENTICATION

Mouse dynamics analyzes the patterns of mouse movement and clicking behavior, offering complementary advantages to keystroke dynamics particularly in contexts where typing is infrequent[1], [10]. This section examines mouse dynamics-based authentication and multimodal systems that combine multiple behavioral signals[8], [11].

A. Segmentation and Feature Extraction for Mouse Dynamics

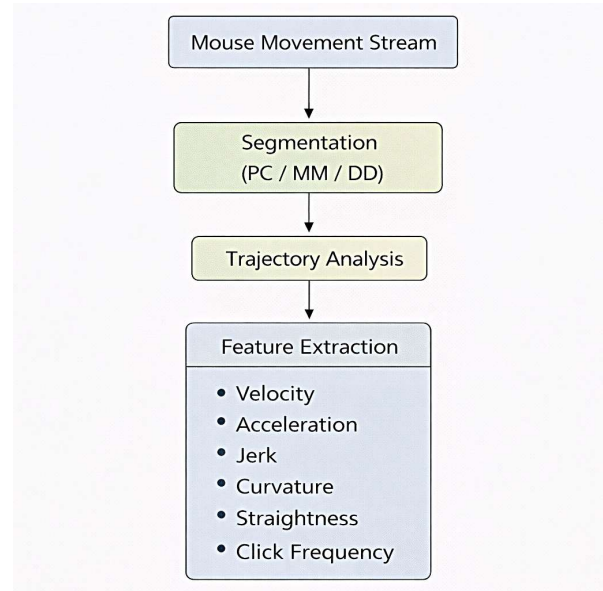


Fig. 5. Mouse movement segmentation and feature extraction pipeline for behavioral authentication.

A critical preprocessing step in mouse dynamics is segmentation, which divides the continuous stream of mouse events into meaningful logical units for feature extraction[1]. Point-and-Click (PC) segmentation identifies sequences of continuous mouse movement events that end with a click event[10]. Mouse-Move (MM) segmentation uses sequences of mouse movement events as segments without requiring a terminating click[1]. Drag-Drop (DD) segments contain sequences starting with a mouse down event, followed by multiple drag events, and concluding with a mouse release event[10].

Kinematic features capture the dynamics of mouse movement. Speed is calculated as the ratio of traveled distance to elapsed time[10]. Velocity captures the rate of position change and can be decomposed into horizontal and vertical components[1]. Acceleration captures the rate of change of velocity[10]. Jerk captures the rate of change of acceleration, providing information about movement smoothness[1]. Geometric features capture the shape and characteristics of movement trajectories. Straightness is the ratio between direct distance and actual trajectory length, with a value of 1 indicating perfectly straight movement[10]. Angle of curvature for any three consecutive points represents direction changes along the path[1].

B. Multimodal Systems Combining Keystroke and Mouse Dynamics

The fusion of keystroke and mouse dynamics into multimodal authentication systems offers substantial

advantages over unimodal approaches[8],[11]. When users interact with computers, they naturally switch between typing and mouse operations depending on the task at hand. A multimodal system can maintain continuous authentication during both typing-intensive tasks and mouse-intensive tasks, providing more consistent coverage[10]. Additionally, the two modalities capture different aspects of user behavior, and their combination can provide more robust and discriminative user models[11].

Research has demonstrated that combining keystroke and mouse dynamics yields better performance than either modality alone[10],[11]. One study found that performance on keystroke data alone was inadequate, while mouse behavior data achieved 85% accuracy through Random Forest, and combining both modalities increased performance to 90%[10]. The SIURUA method specifically addresses authentication in hybrid scenes where users perform diverse tasks by separating scene-related features from user-related features, achieving 84% accuracy in mixed-scenario environments[10].

V. MOBILE SENSOR-BASED AUTHENTICATION

The proliferation of smartphones and tablets equipped with diverse sensors has created new opportunities for behavioral biometric authentication[7],[8]. Modern mobile devices typically include accelerometers, gyroscopes, magnetometers, touchscreens, multiple cameras, GPS receivers, and various environmental sensors, enabling the capture of rich behavioral data[7].

A. Touchscreen Gesture Authentication

Touchscreen gestures represent the mobile analog of mouse dynamics, analyzing finger interaction patterns with capacitive touchscreens[7],[8]. Unlike mouse interactions, touchscreen interactions are direct and can capture pressure, contact area, and multi-touch gestures[5]. Common gestures analyzed include swipes, taps, flicks, pinches, and spreads[7]. Features extracted include velocity, acceleration, path deviation, pressure variation, touch area variation, and gesture duration[11].

B. Motion Sensor Authentication

Motion sensors, particularly accelerometers and gyroscopes, capture device movement and orientation during mobile device use[7]. The accelerometer measures proper acceleration in three spatial dimensions, capturing how users hold and move their devices[8]. The gyroscope measures angular velocity, capturing rotation and orientation changes[7]. Motion sensor-based authentication has been explored in various contexts

including picking up the phone, shaking gestures, in-air handwriting, and free-form gestures[5], [8].

C. Deep Learning for Mobile Authentication

Deep learning approaches have achieved state-of-the-art performance on mobile authentication tasks[11], [15]. A comprehensive evaluation comparing MLP, LSTM, bidirectional LSTM, and convolutional LSTM architectures on a mobile banking dataset found that deep learning models achieved exceptional performance on raw sensor and touch data, with accuracy exceeding 99.85% and FAR, FRR, and EER all below 0.5%[11]. Notably, performance on raw data substantially exceeded performance on handcrafted features, demonstrating the ability of deep networks to learn more discriminative representations directly from raw sensor streams[15].

VI. DEEP LEARNING APPROACHES

Deep learning has revolutionized behavioral biometric authentication by enabling automatic feature extraction from raw data[11], [15] and capturing complex temporal dependencies that traditional machine learning methods cannot represent[13].

A. Convolutional Neural Networks (CNN)

CNNs are particularly effective at extracting local patterns from sequential data[15]. In keystroke dynamics, CNNs can learn characteristic patterns from short sequences of key press events without requiring manual feature engineering[9]. In mouse dynamics, CNNs have been applied to image-based representations of mouse movement trajectories[10], treating the spatial path as an image to be classified[11].

B. Recurrent Neural Networks (RNN, LSTM, GRU)

Recurrent neural networks are specifically designed to handle sequential data[9], [11] and can capture dependencies across time steps. Long Short-Term Memory (LSTM) networks address the vanishing gradient problem that affects standard RNNs[15], enabling them to learn patterns over extended sequences[13]. Gated Recurrent Units (GRUs) provide a computationally efficient alternative to LSTMs with comparable performance[15]. In keystroke authentication, LSTMs can model the temporal evolution of typing patterns across entire sessions[9], [11].

C. Transformer Architectures

Transformer architectures, originally developed for natural language processing[13], have recently been applied to keystroke dynamics and have shown promising

results[13]. Transformers use self-attention mechanisms to capture dependencies between all positions in a sequence simultaneously[13], potentially identifying patterns that sequential RNNs might miss. Recent work has demonstrated that transformer-based models achieve extremely low EER below 0.02% on large-scale keystroke datasets[13].

D. Hybrid Architectures

Hybrid architectures combining multiple types of neural network layers have demonstrated state-of-the-art performance[15], [20]. CNN+RNN architectures use convolutional layers to extract local features followed by recurrent layers to capture longer-range temporal dependencies[15]. CNN+Transformer architectures combine local feature extraction with global attention mechanisms[13].

VII. PRIVACY-PRESERVING AUTHENTICATION

As behavioral biometric authentication systems collect increasingly rich data about user behavior[8],[13], privacy concerns become paramount[12]. Behavioral data can potentially reveal sensitive information including emotional state, health conditions, and demographic characteristics[7], [12].

A. Homomorphic Encryption for Behavioral Data

Homomorphic encryption enables computation on encrypted data without decrypting it[12]. In the context of behavioral authentication, a user can encrypt their behavioral feature vectors before sending them to an authentication server[12]. The server computes similarity scores (such as cosine similarity) on the encrypted data[12] and returns encrypted results, never seeing the raw biometric data[12]. The user then decrypts the results locally to determine authentication outcome. This approach ensures that even if the authentication server is compromised, raw behavioral data remains protected[12].

B. Secure Biometric Template Storage

Biometric templates stored on servers represent a significant security risk if compromised[8], [13]. Unlike passwords, compromised biometric data cannot be changed[8]. Secure template storage techniques including fuzzy commitment schemes and secure sketches enable template storage without exposing raw biometric data[12]. These techniques store only transformed or hashed versions of templates, and authentication is performed in the transformed domain[12].

C. Oblivious Transfer Protocols

Oblivious transfer protocols allow a user to retrieve specific information from a server without revealing which information was retrieved[12]. In behavioral authentication, oblivious transfer can be used to retrieve reference templates for only the specific activities a user performed[12], without revealing to the server which activities those were. This protects user activity privacy even from the authentication server[12].

VIII. BEHAVIORAL DRIFT AND ADAPTIVE AUTHENTICATION

Behavioral drift represents one of the most significant challenges facing behavioral biometric authentication systems[13], [21]. As user behavior evolves over time due to skill acquisition, fatigue, emotional state changes, aging, and device switching[13], static models trained on historical data become progressively less representative[16], leading to increased false rejection rates and degraded user experience[8].

A. Factors Contributing to Behavioral Drift

Skill acquisition is perhaps the most significant factor[13] as users become more experienced with typing, mouse use, or touchscreen interaction, they become faster, more consistent, and develop more efficient patterns. A user's typing speed may increase substantially over months or years of regular computer use[6]. Fatigue and circadian rhythms produce shorter-term fluctuations[21]: users type differently at different times of day, when tired versus alert, and when stressed versus relaxed. Emotional state also influences behavior[18], with studies demonstrating systematic changes in typing patterns under different emotional conditions. Illness and aging produce longer-term changes: neurological conditions affect motor control and typing patterns[13], and general aging slows reaction times. Device and environmental changes cause abrupt shifts[1], [10]: switching from a mechanical desktop keyboard to a laptop keyboard changes key travel distance and tactile feedback[1], affecting hold times and typing rhythm. Similarly, switching from an external mouse to a touchpad alters movement acceleration and trajectory patterns. An adaptive authentication system must implement device-aware modeling that detects such hardware changes[10] and adjusts behavioral profiles accordingly. Additionally, contextual factors including time of day, application type, and user posture (sitting versus standing) significantly influence behavioral patterns[21]. Systems that incorporate time-of-day context adaptation can maintain authentication accuracy by recognizing that a user typing at 9 AM after waking may exhibit different patterns than the same user typing at 9 PM after a full day of work[16].

B. Evidence of Limited Adaptability in Current Systems

A comprehensive survey of keystroke and mouse behavioral biometric systems found that only a small minority of studies (approximately 5-10%) implement any form of adaptation mechanism[13]. The vast majority evaluate authentication performance on data collected within a single session or across sessions separated by days or weeks, without assessing performance over longer periods during which behavioral drift would be expected to occur[2], [8]. This lack of adaptability has important practical consequences: a system that performs well on data collected within a single day may perform poorly when authenticating the same user months later[13], [21].

C. Existing Adaptation Mechanisms

Sliding window techniques maintain a fixed-size window of recent samples[16], discarding older samples as new samples become available, ensuring the user model reflects recent behavior. Growing window techniques accumulate all samples over time[13], retaining all historical information but potentially being slow to adapt to genuine behavioral changes. Transfer learning approaches train models on source data and then fine-tune them on new data from the target session[15], requiring fewer samples to adapt to new conditions. Double serial adaptation strategies use two thresholds to determine when to update user templates[10], adding clearly genuine samples to the reference set.

IX. PROPOSED HYBRID AUTHENTICATION ARCHITECTURE

Based on the comprehensive analysis of existing systems and identified research gaps[13], [8], we propose a Hybrid Adaptive Continuous Authentication (HACA) architecture designed specifically to address the challenges of behavioral drift, contextual variability, and long-term usability[13], [21].

A. Architecture Overview

The proposed architecture consists of three main layers operating in sequence: (1) a sliding window segmentation and multimodal feature extraction layer[16],[10] that processes raw input streams; (2) a hybrid classifier combining LightGBM for rapid classification with One-Class SVM for anomaly detection[20], [6]; and (3) an adaptive profile updating mechanism that incorporates incremental learning to track behavioral evolution[16], [13].

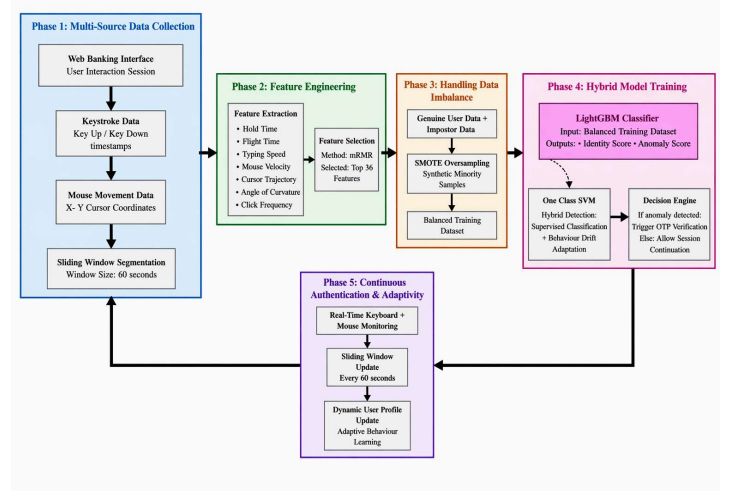


Fig. 6. Proposed hybrid adaptive continuous authentication architecture based on multimodal behavioral biometrics.

B. Layer 1: Sliding Window Segmentation and Feature Extraction

Raw input streams from keyboard, mouse, and (when available) touchscreen and motion sensors are divided into overlapping windows[16]. For keystroke data, each window is processed to extract hold times, flight times, digraph latencies[6], [13], typing speed, and error rates. For mouse data, each window yields movement velocity, acceleration, jerk, curvature, straightness[10], clicking speed, and clicking strength. Contextual features including device type, application in use, time of day[7], [21], and session duration are also extracted. All features are normalized per user[11] to account for baseline differences.

C. Layer 2: Hybrid Classifier with Anomaly Detection

The hybrid classifier employs two complementary mechanisms operating in parallel[14] with distinct roles:

LightGBM serves as the primary supervised classifier[20] for rapid identity verification. Trained on labeled genuine user samples, LightGBM outputs a probability score indicating whether the current behavioral window belongs to the legitimate user[20]. With demonstrated accuracy of 94.68% on keystroke benchmark datasets[20], LightGBM provides fast, reliable classification for routine authentication decisions.

One-Class SVM serves as an anomaly detection boundary validator [6], [18] that operates without requiring impostor training samples[6]. Trained exclusively on legitimate user behavior, the One-Class SVM learns the boundary of normal behavioral patterns[18]. When current behavior

falls outside this learned boundary, it signals potential behavioral drift or impostor activity. This is particularly valuable for detecting novel attacks or behavioral patterns not represented in training data[18].

Decision Fusion: The final authentication decision integrates both outputs using a confidence-weighted mechanism[14]:

- If LightGBM confidence is high (>0.85) and One-Class SVM confirms normalcy, authentication is granted
- If LightGBM confidence is moderate but One-Class SVM indicates anomaly, additional verification is triggered
- If LightGBM confidence is low regardless of anomaly status, authentication is rejected

This dual-classifier approach balances security (anomaly detection catches what supervised learning misses) with usability (LightGBM provides fast, accurate decisions for routine cases)[8].

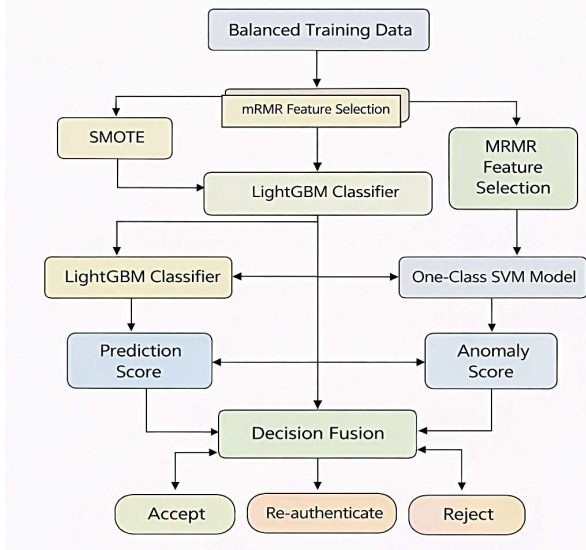


Fig. 7. Hybrid classification framework combining LightGBM prediction with One-Class SVM anomaly detection.

D. Layer 3: Adaptive Profile Updating

When the system detects behavioral drift through sustained anomalies[13] or through verification of genuine user activity following false rejections, the adaptive updating mechanism is triggered. The system collects recent behavioral samples as new training data and performs incremental learning to update the user profile[16]. The update process is carefully controlled to

adapt to genuine behavioral evolution while remaining robust to temporary fluctuations[13]. The update rate is adaptive: during periods of stable behavior, updates are minimal; during periods of rapid behavioral change (such as after device switching)[21], updates are more aggressive.

E. Advantages of the Proposed Architecture

The proposed architecture offers several advantages over existing approaches. First, it explicitly addresses behavioral drift [13] through the combination of short-term classification and long-term anomaly detection with adaptive updating. Second, the hybrid classifier provides both rapid decision-making and consistency monitoring[14], balancing security and usability. Third, the sliding window approach enables real-time operation[16] with minimal latency. Fourth, the architecture is modality-agnostic[8] and can incorporate additional behavioral signals as they become available. Fifth, the adaptive updating mechanism enables the system to evolve with the user over time[13], [16] without requiring complete retraining.

X. CONCLUSION

Behavioral biometric authentication provides a scalable, secure, and user-friendly alternative to traditional authentication methods[8], [13], enabling continuous identity verification throughout active sessions without interrupting user workflow[7], [11]. This survey has provided a comprehensive analysis of keystroke dynamics, mouse dynamics, touchscreen gestures, and sensor-based behavioral signals, along with the machine learning and deep learning techniques employed for identity verification[11], [15].

Key findings from this survey include: (1) deep learning approaches, particularly LSTM and hybrid CNN+RNN architectures, achieve state-of-the-art performance on behavioral authentication tasks[15], with accuracy exceeding 99% on mobile sensor data[11]; (2) gradient boosting methods such as LightGBM achieve exceptional performance on keystroke dynamics[20], reaching 94.68% accuracy on benchmark datasets[20]; (3) multimodal fusion combining keystroke and mouse dynamics improves robustness and accuracy compared to unimodal systems[10], [11]; (4) behavioral drift due to skill acquisition, fatigue, device switching, and aging remains a significant challenge[13], [21], with only a small minority of current systems implementing adaptation mechanisms[13]; (5) privacy-preserving techniques

including homomorphic encryption enable secure authentication without exposing raw behavioral data[12].

Future systems should integrate contextual awareness to account for task, device, and environmental variations[21]; incorporate adaptive learning mechanisms to track behavioral evolution over extended periods[16]; adopt privacy-preserving architectures to protect sensitive behavioral data[12]; and leverage multimodal fusion to improve robustness across diverse usage scenarios[10]. The hybrid adaptive authentication architecture proposed in this survey provides a framework for addressing these requirements, combining real-time classification with anomaly detection and incremental profile updating to maintain authentication accuracy as user behavior naturally evolves over time[16], [20].

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